

The Robot & the Crystals



A robot is traveling through an energy tunnel made of N stations arranged in a straight line.

Each station contains an energy crystal:

- Positive values increase the robot's energy
- Negative values decrease it

The robot:

- starts at station **1**
- may move:
 - 1 station forward
 - or 2 stations forward (skips one station)

Whenever the robot lands on a station, it collects that station's energy value.

The robot also has a special power:

During the journey, it may choose **at most one visited station** whose energy value will not be counted.

The robot continues until it reaches the end of the tunnel (beyond all the stations).

Input Format

You are given:

- An integer N — number of stations
 - An array:
 - $A[1..N]$ — energy value at each station
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Rules

- The robot starts at station 1
 - From station i , the robot may move:
 - to $i + 1$
 - or to $i + 2$
 - The robot collects the value of every station it visits
 - The robot may ignore the value of **at most one visited station**
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Objective

Determine the **maximum total energy** the robot can collect.

Example

Input

$N = 6$

$A = [3, -5, 4, -2, 7, 1]$

Possible Paths

1- Path:

$$1 \rightarrow 3 \rightarrow 5 \rightarrow 6$$

Collected energy:

$$3 + 4 + 7 + 1 = 15$$

No skip needed.

Final energy:

$$15$$

2- Path:

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 6$$

Collected energy:

$$3 + (-5) + (-2) + 1 = -3$$

Use the special skip ability on station 2:

$$3 + 0 + (-2) + 1 = 2$$

Final energy:

$$2$$

3- Path:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 5$$

Collected energy:

$$3 + (-5) + 4 + 7 = 9$$

Use skip on station 2:

$$3 + 0 + 4 + 7 = 14$$

Final energy:

$$14$$

Optimal Choice

The best path is:

$$1 \rightarrow 3 \rightarrow 5 \rightarrow 6$$

Maximum obtainable energy:

15

Additional Test Cases

Test Case 1

Input

6
3 -5 4 -2 7 1

Output

15

Test Case 2

Input

5
1 -10 2 3 4

Explanation

Best path:

$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Collected energy:

$1 + 2 + 3 + 4 = 10$

Output

10

Test Case 3

Input

7
5 -8 6 -1 -2 10 -3

Explanation

Best path:

$1 \rightarrow 3 \rightarrow 4 \rightarrow 6$

Collected energy:

$$5 + 6 + (-1) + 10 = 20$$

Use skip on station 4:

$$5 + 6 + 0 + 10 = 21$$

Output

21

Test Case 4

Input

4

-1 -2 -3 -4

Explanation

Robot lands on 1st station, collects energy at station 1, then skips 1 station and lands on station 3 and ignores its value and then skips over the last station.

Collected energy:

-1

Use skip on station 1:

Output

-1

Hint:

First think of this problem without using the ignore station value option. Then once you figure it out, add the ability to ignore the value of one station to your solution.

Here is a reminder of the outline of your solution

Part 1: Divide & Conquer

1. Define the value returned by the function f which we want to optimize.
 2. Define the parameters which f depends on.
 3. Draw the recursion tree for f using the values from the example above.
 4. Write the recursive (divide and conquer) code to solve the question.
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Part 2: Dynamic Programming

5. Draw the table and determine the dependencies between the table cells.
 6. Determine the direction of movement within the table.
 7. Write the Dynamic Programming code which fills the table(s).
 8. Write the code that prints the sequence of moves leading to the optimal solution.
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Submission Instructions

You are requested to submit a report on Moodle that explains each of the steps above. The report should also include graphs and figures to explain your solution and the rationale behind it.

In addition, you must submit your working code on the Hackerrank website as usual. The link will be provided on Moodle / Facebook Group.

Important

To solve this task, you are **not allowed** to copy or be inspired by any piece of code from:

- the internet
- AI tools
- colleagues
- or any external source

If any resemblance is found between your code and code available online or generated by AI (even if written for a different problem), it will be considered cheating.

You are only allowed to review:

- the three problems solved in class
- the previous years' problems posted on Moodle

If it is proven that you cheated on this task (regardless of how small the similarity percentage is), you will receive **zero in the final exam mark**.

Good Luck

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